STACKS, MODULI OF CURVES: LEMMA-STABLE-QUASI-COMPACT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3503-3506 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 3508-3554, the complete author proof (45 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 12 of queue_moduli-curves.json (pb=1491, stmt 3503-3506, proof 3508-3554).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BS6: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3554–3554. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E8X: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3554–3554. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPZ: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 420–424. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E96: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3554–3554. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPY: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 370–402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $g \geq 2$. The stack $\overline{\mathcal{M}}_g$ is quasi-compact.*

Date: stacks-project:moduli-curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We will use the notation from Section 0DPY. Consider the subset

$$T \subset |PolarizedCurves|$$

of points ξ such that there exists a field k and a pair $(X, \mathcal{L})$ over k representing ξ with the following two properties

(1) X is a stable genus g curve, and
(2) $\mathcal{L} = \omega_X^{\otimes 3}$.

Clearly, under the continuous map

$$|PolarizedCurves| \longrightarrow |\mathcal{C}urves|$$

the image of the set T is exactly the open subset

$$|\overline{\mathcal{M}}_g| \subset |\mathcal{C}urves|$$

Thus it suffices to show that T is quasi-compact. By Lemma 0DPZ we see that

$$|PolarizedCurves| \subset |\mathcal{P}olarized|$$

is an open and closed immersion. Thus it suffices to prove quasi-compactness of T as a subset of $|\mathcal{P}olarized|$. For this we use the criterion of Moduli Stacks, Lemma 0E96. First, we observe that for $(X, \mathcal{L})$ as above the Hilbert polynomial P is the function $P(t) = (6g-6)t + (1-g)$ by Riemann-Roch, see Algebraic Curves, Lemma 0BS6. Next, we observe that $H^1(X, \mathcal{L}) = 0$ and $\mathcal{L}$ is very ample by Algebraic Curves, Lemma 0E8X. This means exactly that with $n = P(3) - 1$ there is a closed immersion

$$i : X \longrightarrow \mathbf{P}^n_k$$

such that $\mathcal{L} = i^*\mathcal{O}_{\mathbf{P}^1_k}(1)$ as desired. □